

Networking Fundamentals & Hands-on Checks

- ◇ Map the OSI Vs TCP/IP models in your own words
- ◇ Run the essential connectivity commands
- ◇ Capture a mini network check for a target host/service

The OSI model is a conceptual framework for understanding how data is transmitted across a network, and it consists of seven layers: application, presentation, session, transport, network, data link physical layer.

The TCP/IP model consists of an Application, Transport, Internet, and network access layer.

srno	OSI model	TCP/IP model
1	Physical: this layer concerns the actual cables and wires that connect to computers, as well as the signals that pass through the wires.	Network Access/link: combines OSI's Physical + Data link. Handles hardware addressing and physical transmission.
2	Data link: this protocol deals with communication between devices that are directly connected to each other on the same network.	Internet: This is similar to OSI's network layer. Ip routing occurs here.
3	Network: Routes data to different networks using IP addresses to find the most efficient way for packets to travel.	Transport: Same OSI's Transport. It offers two transport protocols: TCP and UDP.
4	Transport: Ensures reliable (or unreliable) delivery between applications. Manage segmentation, flow control, and error recovery (e.g., TCP, UDP).	Application: Merges OSI's session + presentation + application. All user-facing protocols and services.
5	Session: manage session establishment, synchronization, and termination between different applications.	
6	Presentation: Translate data, handles encryption/decryption,	

	and compression. Ensure that data is readable by the application.	
7	Application: The layer at which users interact. Provides network services at the application level, eg: HTTP, FTP, SMTP	

- ◇ Key differences:
- ◇ The OSI model is theoretical, whereas the TCP/IP model is practical.
- ◇ OSI has more detailed layering (7 levels instead of 4), which is helpful for explaining theory.
- ◇ The TCP/IP protocol mirrors the current reality of the internet.
- ◇ Quick Concepts
- ◇ OSI layer (L1-L7) vs TCP/IP stack (Link, Internet, Transport, Application)

srno	OSI Model	TCP/IP
1	L1: Physical layer transfer of raw data via a medium (cables, radio waves, voltages) is the subject of this study. Specifies hardware parameters such as signal timing and connectors.	L1: Link integrates the data link as OSI Physical layers. Manages local network hardware.
2	L2: Data connection; it manages error detection and correction in a local network and facilitates node-to-node data transfer. Mac addresses are used to identify devices.	L2: Internet layer comparable to the OSI Network layer in charge of IP protocol logical addressing and routing.
3	L3: Network: Manages packet routing and logical addressing across various networks. IP addresses are used to determine the most efficient data delivery path.	L3: Transport is similar to the OSI transport layer for end-to-end communication via TCP or UDP provided.
4	L4: Transport oversees reliability, flow control, and end-to-end communication. Offers	L4: Application is similar to OSI session; presentation, and application layers are combined. All high-level


```
ubuntu@ip-172-31-42-193:~$ hostname -I
172.31.42.193
```

- Reachability: ping <target> - mention latency and packet loss.
 - Ping trainwithshubham.com, latency: min = 0.543 ms , average = 0.587 ms, maximum = 0.649 ms, mdev = 0.036 ms. There is no loss of packet.

```
ubuntu@ip-172-31-42-193:~$ ping -c 5 trainwithshubham.com
PING trainwithshubham.com (3.33.251.168) 56(84) bytes of data.
64 bytes from aec037177372cc6cd.awsglobalaccelerator.com (3.33.251.168): icmp
_seq=1 ttl=246 time=0.552 ms
64 bytes from aec037177372cc6cd.awsglobalaccelerator.com (3.33.251.168): icmp
_seq=2 ttl=246 time=0.556 ms
64 bytes from aec037177372cc6cd.awsglobalaccelerator.com (3.33.251.168): icmp
_seq=3 ttl=246 time=0.568 ms
64 bytes from aec037177372cc6cd.awsglobalaccelerator.com (3.33.251.168): icmp
_seq=4 ttl=246 time=0.583 ms
64 bytes from aec037177372cc6cd.awsglobalaccelerator.com (3.33.251.168): icmp
_seq=5 ttl=246 time=0.594 ms

--- trainwithshubham.com ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4114ms
rtt min/avg/max/mdev = 0.552/0.570/0.594/0.015 ms
```

- Path: traceroute <target> (tracpath) - note any long hops/ timeouts.
 - Traceroute trainwithshubham.com there are two long hops hop 4 5.044 ms and hop 8 5.745 ms. there is complete blackout after 8 till 30, 22 timeouts.

```
ubuntu@ip-172-31-42-193:~$ traceroute trainwithshubham.com
traceroute to trainwithshubham.com (15.197.225.128), 30 hops max, 60 byte packets
 1 244.5.0.21 (244.5.0.21) 1.550 ms 244.5.0.13 (244.5.0.13) 3.405 ms 244.5.0.23 (244.5.0.23) 5.892 ms
 2 * * 100.65.16.128 (100.65.16.128) 1.107 ms
 3 100.66.8.68 (100.66.8.68) 13.393 ms 100.66.9.110 (100.66.9.110) 7.885 ms 100.66.8.92 (100.66.8.92) 2.220 ms
 4 100.66.11.38 (100.66.11.38) 1.446 ms 100.66.10.70 (100.66.10.70) 1.070 ms 100.66.11.130 (100.66.11.130) 1.050 ms
 5 240.4.192.5 (240.4.192.5) 0.207 ms 240.4.192.6 (240.4.192.6) 0.324 ms 0.313 ms
 6 240.3.120.13 (240.3.120.13) 1.240 ms 1.573 ms 1.969 ms
 7 240.3.120.31 (240.3.120.31) 0.848 ms 240.3.120.18 (240.3.120.18) 0.648 ms 240.3.120.30 (240.3.120.30) 0.670 ms
 8 242.8.228.117 (242.8.228.117) 5.508 ms 242.8.228.245 (242.8.228.245) 5.033 ms 242.8.229.245 (242.8.229.245) 5.172 ms
 9 * * *
10 * * *
11 * * *
12 * * *
13 * * *
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30 * * *
```

- Ports: `ss -tulpn` (or `netstat -tulpn`) -- list one listening service and its port.
 - `Netstat -tulpn`: multiple entries for the same port: there are multiple entries for port 22, 80, 5453, 323. This suggests that multiple processes or services might be listening to those ports.

```
ubuntu@ip-172-31-42-193:~$ sudo netstat -tulpn
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 127.0.0.53:53          0.0.0.0:*               LISTEN      338/systemd-resolve
tcp        0      0 0.0.0.0:22            0.0.0.0:*               LISTEN      1/init
tcp        0      0 0.0.0.0:80            0.0.0.0:*               LISTEN      620/nginx: master p
tcp        0      0 127.0.0.54:53         0.0.0.0:*               LISTEN      338/systemd-resolve
tcp6       0      0 :::22                 :::*                   LISTEN      1/init
tcp6       0      0 :::80                 :::*                   LISTEN      620/nginx: master p
udp        0      0 127.0.0.54:53         0.0.0.0:*               338/systemd-resolve
udp        0      0 127.0.0.53:53         0.0.0.0:*               338/systemd-resolve
udp        0      0 172.31.42.193:68      0.0.0.0:*               487/systemd-network
udp        0      0 127.0.0.1:323         0.0.0.0:*               691/chronyd
udp6       0      0 :::1:323              :::*                   691/chronyd
```

- Name resolution: `dig <domain>` or `nslookup <domain>` - record the resolved IP
 - `Nslookup trainwithshubham.com`: the resolved IP 3.33.251.168, 15.197.225.128.

```
ubuntu@ip-172-31-42-193:~$ nslookup trainwithshubham.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   trainwithshubham.com
Address: 3.33.251.168
Name:   trainwithshubham.com
Address: 15.197.225.128
```

- HTTP check `curl -I <http/https -url>` - note the HTTP status code
 - The HTTP status code is 405.

```
ubuntu@ip-172-31-42-193:~$ curl -I trainwithshubham.com
HTTP/1.1 405 Not Allowed
Server: awselb/2.0
Date: Fri, 13 Feb 2026 10:56:19 GMT
Content-Length: 0
Connection: keep-alive
WAFRule: 0
```


- Connections snapshots: `netstat -an | head` – count ESTABLISHED vs LISTEN (rough).
 - `Netstat -an | head` there are three ESTABLISHED active connections to the other host.

```
ubuntu@ip-172-31-42-193:~$ netstat -an | head
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp    0      0 127.0.0.53:53           0.0.0.0:*               LISTEN
tcp    0      0 0.0.0.0:22              0.0.0.0:*               LISTEN
tcp    0      0 0.0.0.0:80              0.0.0.0:*               LISTEN
tcp    0      0 127.0.0.54:53           0.0.0.0:*               LISTEN
tcp    0      0 172.31.42.193:22        1.38.144.12:30885       ESTABLISHED
tcp    0      0 172.31.42.193:22        1.38.144.12:61107       ESTABLISHED
tcp    0      0 172.31.42.193:22        103.162.47.202:56539    ESTABLISHED
tcp    0      0 172.31.42.193:22        1.38.144.12:61111       ESTABLISHED
```

- LISTEN: These are ports waiting for incoming connections.
- ❖ Mini Task: Port Probe & Interpret
 - Identify one listening port from `ss -tulpn` (eg ssh on 22 or a local web app).
 - Ssh: Listening on port 22.
 - Local web application: Listening on port 80.

```
ubuntu@ip-172-31-42-193:~$ ss -tulpn
Netid State     Recv-Q Send-Q           Local Address:Port      Peer Address:Port      Process
udp    UNCONN    0      0           127.0.0.54:53           0.0.0.0:*
udp    UNCONN    0      0           127.0.0.53%lo:53        0.0.0.0:*
udp    UNCONN    0      0       172.31.42.193%ens5:68    0.0.0.0:*
udp    UNCONN    0      0           127.0.0.1:323           0.0.0.0:*
udp    UNCONN    0      0           [::]:323                [::]:*
tcp    LISTEN    0      4096       127.0.0.53%lo:53        0.0.0.0:*
tcp    LISTEN    0      4096           0.0.0.0:22             0.0.0.0:*
tcp    LISTEN    0      511           0.0.0.0:80             0.0.0.0:*
tcp    LISTEN    0      4096       127.0.0.54:53          0.0.0.0:*
tcp    LISTEN    0      4096           [::]:22                [::]:*
tcp    LISTEN    0      511           [::]:80                [::]:*
```

- From the same machine, test it: `nc -zv localhost <port>` (or `curl -I http://localhost:<port>`).
 - `Nc -zv localhost 80`: connection to localhost (127.0.0.1) 80 port [tcp/http] succeeded!

```
ubuntu@ip-172-31-42-193:~$ nc -zv localhost 80
Connection to localhost (127.0.0.1) 80 port [tcp/http] succeeded!
```

- Write one line: is it reachable? If not, what is the next check? (eg: service status, firewall).

- Yes, it is reachable. Connection to localhost (127.0.0.1) 80 port [tcp/http] succeeded!

❖ Reflection (add to your markdown)

- Which command gives you the fastest signal when something is broken?
 - Ping command generally gives the fastest signal when something is broken.
- What layer (OSI/TCP -IP) would you inspect next if DNS Fails? If HTTP 500 shows up?
 - If DNS Fails
 - TCP/IP: layer 7 (Application)
 - Reasons:
 - ◆ DNS is an application-layer protocol
 - ◆ Common issues include resolver misconfiguration, DNS service failure, or invalid record.
 - ◆ If unresolved, move to: Transport layer.
- Two follow up checks you had run in a real incident.
- DNS Troubleshooting:
 - Cat /etc/resolv.conf
 - Dig trainwithshubham.com
- HTTP 500 Troubleshooting:
 - Tail -f application.log
 - Systemctl status trainwithshubham.com